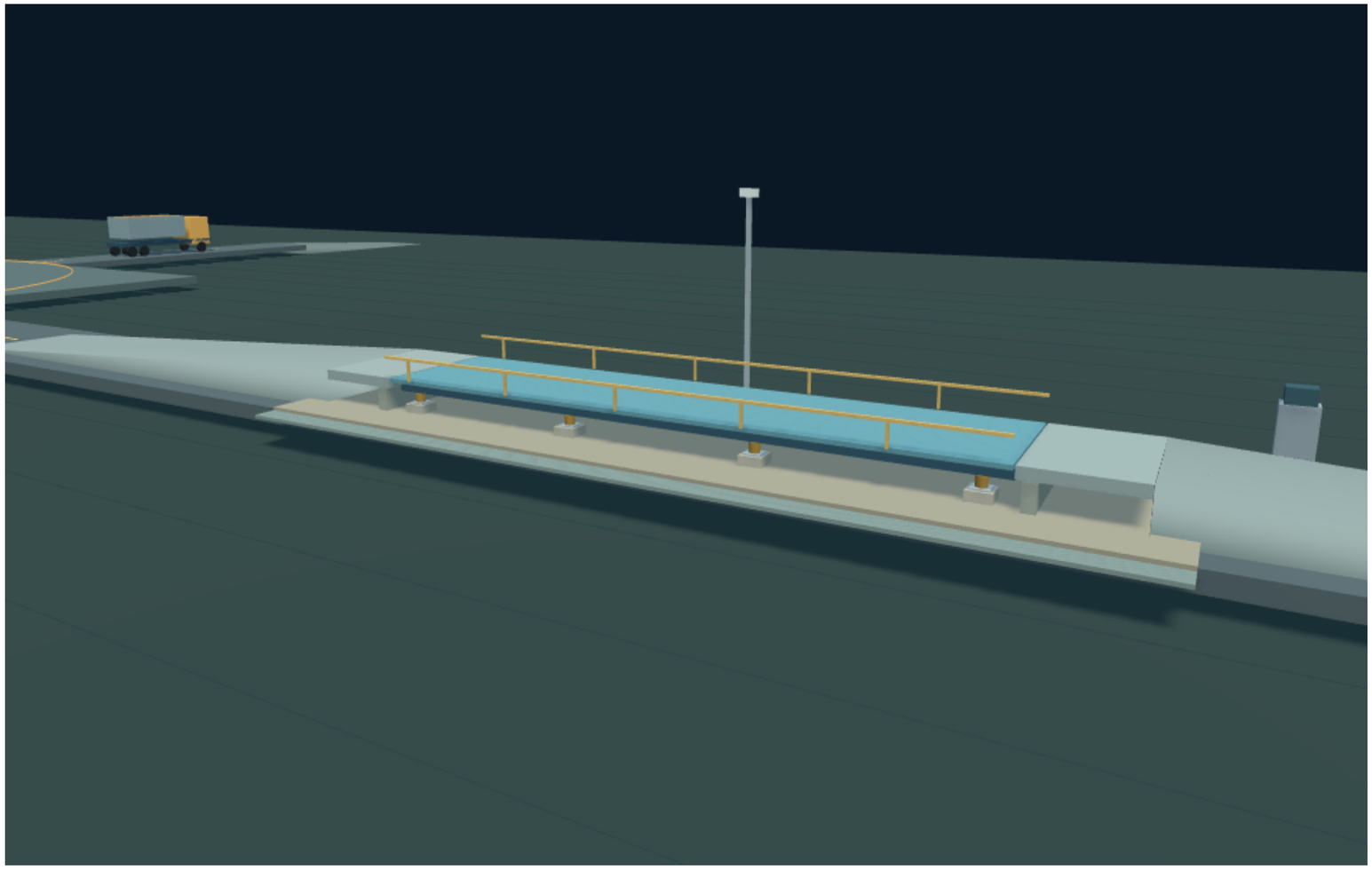
AIMOC WIM

Scenario 02 - Static inbound / WIM outbound

CONFIDENTIAL - CONCEPTUAL



INBOUND: +X to -X, Z = -65 m

OUTBOUND: +Z to -Z, X = 0

Pitless deck 20 x 4.5 m at +2.10 m; rise 1.20 m

Static ramps 2 x 24 m at 5%; landings 2 x 3 m

Static footprint 74 m; shallow raft subject to geotech

Rod gap 50 mm; lane edge 225 mm; pad edge 1225 mm

Canopy 40 x 8 m; clear 7 m; gantry 6.5 m

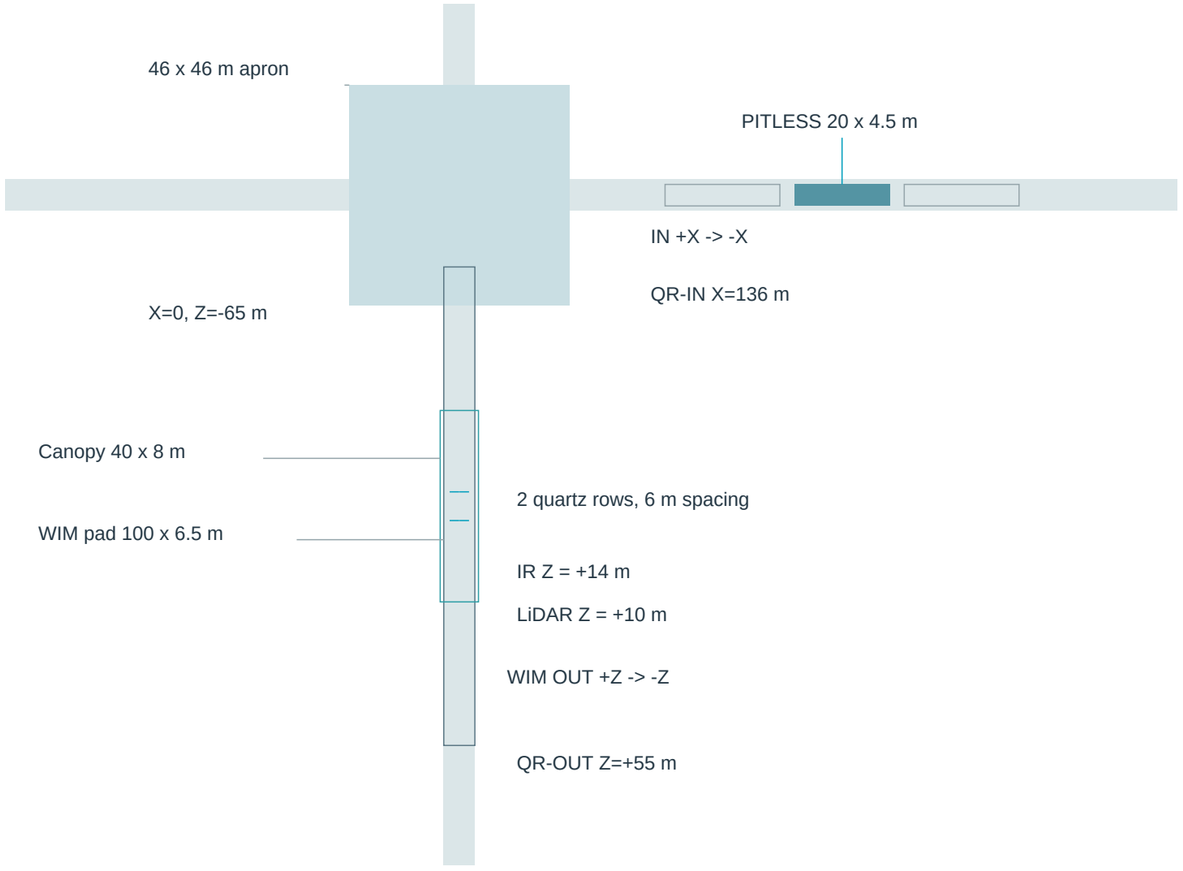
Intersection apron 46 x 46 m; outer radius 23 m

Flow: Oracle NetSuite -> entry QR -> stable static tare -> loading -> exit QR -> WIM gross -> AIMOC -> ERP.

Order differences are displayed numerically only. No match / mismatch decision.

QR does not replace infrared. The single-truck animation has no queue; traffic and swept-path analysis are required.

Not construction drawings. Dimensions/capacities require OEM/civil/metrology verification.



CIVIL / STRUCTURE

Yard +0.90 m; static deck +2.10 m; rise 1.20 m

WIM: concrete 450 mm + base 300 mm + subbase 150 mm

Power: panel -> SPD -> UPS -> controller / indicator / gateway.

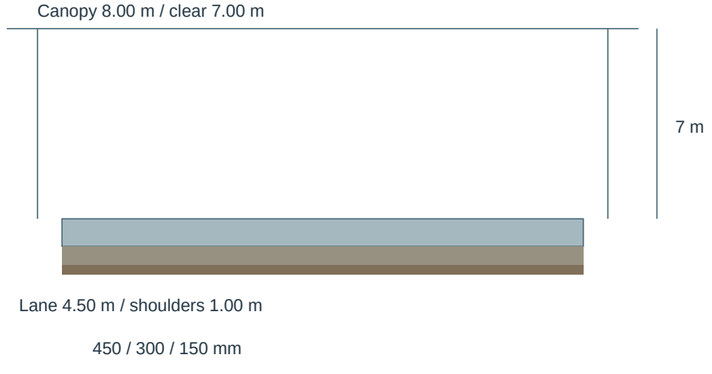
Data: indicator RS-232/485/Ethernet -> IoT gateway -> LAN/fiber -> AIMOC edge -> cloud -> Oracle NetSuite middleware.

Separate power/data; sleeved crossings. PoE CCTV; 4G failover, optional WLAN.

Horizontal/existing earth system, flexible bonding, SPD and lightning protection; raised/shallow ducts. Sizes from MEP study.

Stages: survey/geotechnics -> shallow preparation -> platform/raft/pedestals -> ramps -> curing/strength test -> installation -> calibration/UAT.

WIM / CANOPY CROSS-SECTION



Not construction drawings. Dimensions/capacities require OEM/civil/metrology verification.

Component	Qty	Description
Digital static weighbridge	1	Raised pitless deck + load cells / mounts + junction box · 20 × 4.5 m · rise 1.2 m · 120 t concept; OEM verification
Digital weight indicator	1	Stable flag + RS-232/RS-485/Ethernet; protocol must be confirmed
Static IoT gateway / modem	1	Protocol adapter, edge buffer, Ethernet + 4G failover
2D QR scanner	2	Entry + exit · covered kiosk; stop / slow presentation
Quartz WIM strip	4	2 rows × 2 strips × 2 m · gap 50 mm
WIM controller	1	4 quartz channels · local processing
Infrared separator	1	1 Tx/Rx set = 2 posts; vehicle separation, not identity
LiDAR	1	Outbound contour · dimensions only
Fixed CCTV	2	1 entry + 1 exit · evidence only; no ANPR/PTZ
Cabinets / UPS / LAN	2	Static and WIM nodes · surge protection / grounding
Traffic signals & interlock	1	Radar/vehicle zone detection + fail-safe controller; not QR alone

Cost: fewer WIM devices do not automatically guarantee total savings. Static deck, indicator, foundations and gateway must be quoted.

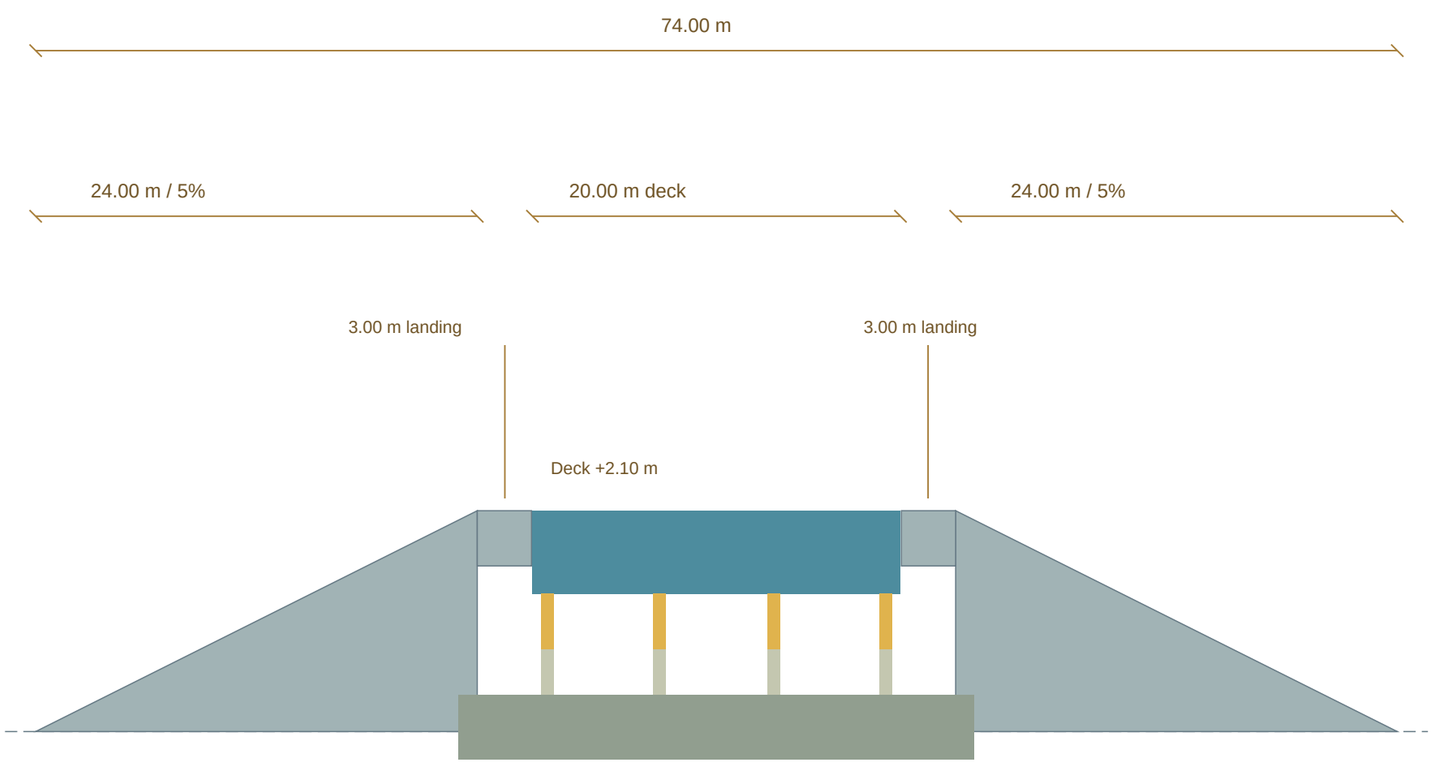
Trip: HYB-4dd36462-6504-46f3-aae2-e696c17070c1 | Oracle NetSuite order: NS-SO-DEMO-2026-001

Tare: 23100 kg | Gross: 71100 kg | Net: 48000 kg | Volume: 32 m3

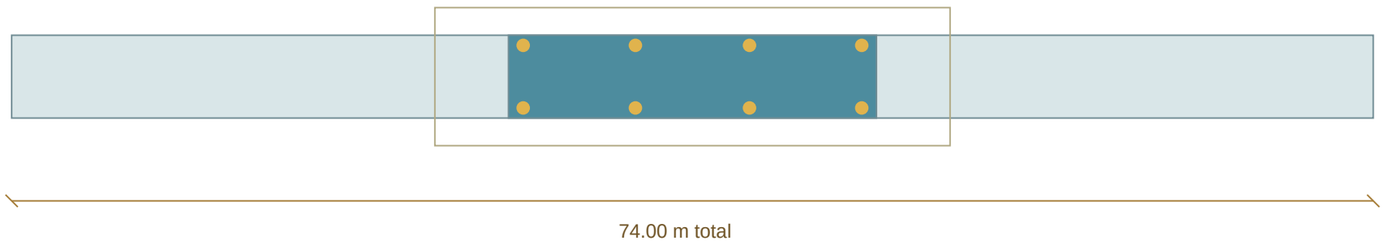
JSON is synthetic; Oracle NetSuite/ERP is not connected. Preserve trip/order IDs, timestamp, weight sources and idempotency key. Invalid weights are not finalized.

Not construction drawings. Dimensions/capacities require OEM/civil/metrology verification.

LONGITUDINAL SECTION (vertical scale exaggerated)



DETAIL PLAN (metres)



Deck: +2.10 m | yard: +0.90 m | rise: 1.20 m

Deck depth: 450 mm* | underside: +1.65 m

Raft top: +1.10 m | bottom: +0.75 m

Pedestal/grout: 250 mm* | mount: 300 mm*

Under-deck zone: 550 mm* (not personnel access)

Preparation: 150 mm* below yard datum

8 support points* | side service band 1.00 m

End gap: 50 mm*; final clearance per OEM

Static deck: X=70...90 m, Z=-65 m

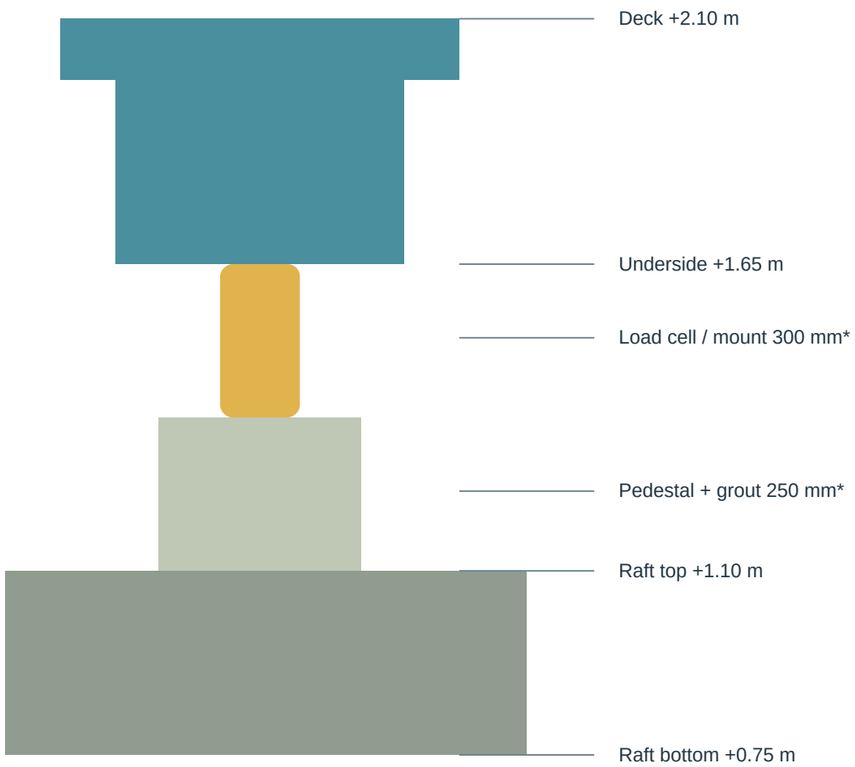
Ramps + landings: X=43...117 m

Entry QR: X=136 m | stop line: X=28 m

*Visual assumptions, not shop drawings. Foundation and capacity require geotechnical, structural and OEM acceptance.

Not construction drawings. Dimensions/capacities require OEM/civil/metrology verification.

SUPPORT DETAIL - NOT AN ORDINARY SPRING



Foundation concept

Shallow raft/beam-slab on a verified platform. Nominal preparation 150 mm below yard datum; not an excavation permit limit or proof of adequate ground.

Support points

8 points are illustrative only. OEM determines count, location, cell ratings, eccentric reactions, dead load and dynamic allowances. Deck capacity 120 t remains a concept target.

Reclaimed ground

Assess stratigraphy, bearing, consolidation, differential settlement, groundwater and liquefaction as applicable. Select a shallow raft only after geotechnical acceptance. Geogrid/aggregate does not eliminate deep soft-soil consolidation.

If shallow foundations are unsuitable

Compare relocation, preload/PVD, stabilization, lightweight materials or permitted deep support. Do not force shallow construction as a guaranteed solution; study methods, permits, time and cost.

Load path

Tyres -> deck -> mounting/load cell -> pedestal/grout -> raft -> verified ground. No ordinary springs; no deck contact with ramps, rigid services or debris.

Maintenance access

1 m side service band reserved. The 0.55 m under-deck zone is not personnel access; service from the side, with OEM jacking points, isolation and mechanical supports.

Not construction drawings. Dimensions/capacities require OEM/civil/metrology verification.

Ramps and landings

Two 24 m ramps at 5%; 3 m level landing each side. Total straight footprint 74 m. Check breakover, overhang, ground clearance, traction and transition settlement against OEM truck data.

Water, tidal flooding and corrosion

Yard +0.90 m and deck +2.10 m are not proven flood-safe levels. Establish design flood level plus freeboard. Use surface/gravity drainage to an approved outfall; check backwater. Marine coating, compatible fixings/materials and sealed glands.

Required survey inputs

Reclamation age/material/compaction history; independent benchmark outside moving ground; soil profile and groundwater; consolidation/settlement data; utilities/permits; topography, flood/tide levels and outfall; fleet wheelbase, overhang and clearance; OEM support reactions/dead load.

MEP without deep trenches

Use raised trays/plinths or approved shallow routes; electrical engineer designs/tests horizontal or existing earth systems. An excavation limit alone neither permits nor prohibits deep rods/piles. Flexible bonding; no welding current through load cells.

Instrumentation and transactions

Load cells -> junction box/indicator -> stable flag -> gateway -> AIMOC. Tare requires all wheels on deck, no external contact, stopped vehicle, stable reading and correct QR identity. Verify statutory acceptance for the instrument classification.

MEP design needs project data

Do not size cables, breakers, earthing resistance targets, drainage flow, pumps or coating thickness from this concept alone. Soil measurements, load schedules, corrosion exposure and equipment protection ratings inform the design. A shallow earth system on reclaimed soil is not guaranteed to meet targets without testing.

Responsibility split

Geotechnical: ground acceptance, monitoring and settlement limits. Civil/structural: foundations, ramps, drainage and loading release. OEM: deck, mounts, load cells, reactions and tolerances. AIMOC: sensor/indicator coordination, connectivity, platform and integration; civil work follows the construction contractor scope. Authority: applicable approvals and statutory verification.

Not construction drawings. Dimensions/capacities require OEM/civil/metrology verification.

Concept sequence: investigation/permits -> design acceptance -> shallow/platform preparation -> ground QC -> raft/pedestals -> curing and strength tests -> ramps/landings -> deck/mounts -> cables/bonding/drainage -> calibration/verification -> Oracle NetSuite/QR integration -> UAT -> settlement monitoring. No curing duration or foundation release date is guaranteed without material data and test results.

HP1 - Survey & design

Independent benchmark, existing utilities, excavation permits, flood/tide levels, permitted CPTu/SPT/lab data; geotechnical acceptance before foundations.

HP2 - Platform & ground

Material/compaction QC to geotechnical specification; verify bearing and settlement. No single generic CBR/Proctor target for all layers.

HP3 - Concrete & supports

Check levels, dimensions, reinforcement, cover, anchors, curing and strength results; designer authorizes loading, not calendar age alone.

HP4 - Mechanical installation

Grout, mounting alignment, clearance, checks/stops, side access and flexible cables. No post-installation drilling/welding without OEM procedures.

HP5 - Calibration & verification

Zero, repeatability, eccentric/corner loading, load range, stable flag, QR trip, API and sensor/power/network faults; procedures/tolerances from OEM/authority.

HP6 - Reclamation monitoring

Baseline before loading; survey every support/transition after commissioning and at risk-based intervals. Geotechnical engineer/OEM set alarm/action limits; re-level and recalibrate as required.

Do not finalize weight with instability, binding, wheels on a ramp, failed load cells or settlement beyond accepted limits. Mechanical/level corrections require zero checks and calibration/statutory verification as applicable.

T01 | Rice Lake Weighing Systems | Truck Scale Foundations - technical article

Open official reference

Pitless selection, service access, drainage and OEM foundation compatibility. Example bearing pressures are not adopted as Marunda soil data.

T02 | FHWA | Ground Improvement / Geotechnical Engineering Circular 13

Open official reference

Ground-treatment selection and performance-verification framework; no site-specific method is selected.

T03 | FHWA-HRT-15-080, 2016 | Synthesis and Evaluation of the Service Limit State of Engineered Fills for Bridge Support, Ch. 1

Open official reference

Settlement and foundation-to-approach transitions. Bridge research is a geotechnical analogy, not a ready-made scale foundation detail.

T04 | BSN official catalogue | SNI 8460:2017 - Persyaratan perancangan geoteknik

Open official reference

Standard identity/scope checked in the catalogue. Full licensed text was not reviewed; the designer must check applicable requirements and edition.

T05 | BSN official catalogue | SNI 2847:2019 - Persyaratan beton struktural

Open official reference

Concrete-design reference subject to structural-engineer applicability review. Concept thicknesses are not claimed to satisfy the standard.

T06 | OIML | OIML R 76-1:2006 - Non-automatic weighing instruments

Open official reference

Requirements/testing reference for applicable non-automatic instruments. QR automation does not establish classification or Indonesian approval.

Document status

Concept study based on open references in the original technical version. This ERP naming revision only aligns Oracle NetSuite terminology; the engineering concept is unchanged and no new OEM or site validation is included. Not signed structural design, geotechnical results, metrology approval or brand selection. SNI identity was checked in official catalogues; licensed text was not reviewed in full. Starred dimensions and all acceptance criteria must be agreed by OEM, geotechnical, structural, MEP and owner teams before construction.